

Tanjil Sarker Rafi

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SUMMARY

Computing Science student at the University of Alberta with software development internship experience building and maintaining full-stack features on an Agile team. Comfortable across front-end (React.js, Next.js, TypeScript) and back-end work, REST APIs, SQL and NoSQL databases, and Git-based collaboration, with a strong object-oriented programming foundation in Java. Fluent in English and French.

EXPERIENCE

- Software Developer Intern** May 2024 – September 2024
 - Nexfolyo (Hybrid), Edmonton, AB*
 - Developed and maintained full-stack features, building user-facing front-end components in React.js on a 6–7 person Agile team.
 - Reviewed peers' code through pull requests to ensure consistency and adherence to best practices.
 - Troubleshooted and resolved software issues, coordinating changes through Git and GitHub using branches and pull requests.
 - Delivered work against requirements reviewed weekly with a product manager, presenting written and verbal progress updates.
- Teaching Assistant, CMPUT 174 (Introductory Python Programming)** Sep 2025 – Dec 2025
 - University of Alberta, Edmonton, AB*
 - Evaluated student code and explained correct approaches clearly to students at widely different skill levels.

EDUCATION

- University of Alberta** Expected July 2027
 - Bachelor of Science, Major in Computing Science, Minor in Business* Edmonton, AB
- Relevant Coursework:** Introduction to Software Engineering; Web Applications and Architecture; File and Database Management; Search and Planning in AI; Machine Learning; Applied Statistics I and II
- Awards:** University of Alberta International Admission Scholarship; Regional Excellence Scholarship

PROJECTS

- BeefStock — Paper-Trading Education Platform** React, TypeScript, Express, Supabase
 - Built a full-stack platform teaching market mechanics through RSI/MACD/SMA indicator math and \$100,000 in risk-free paper trading, with no real money or personalized buy/sell advice.
 - Architected a shared types layer between the React front end and Express API, plus an LLM-ready chatbot seam with safety guardrails enforced on every data and chat screen.
- SpecAQI — Hyperlocal Air Quality Tracker** Next.js, React, Tailwind CSS, Leaflet
 - Built at the HackED Hackathon (University of Alberta): a full-stack web app reporting air quality for a specific place — a park, beach, campsite, or mall — rather than a city-wide average.
 - Integrated the PurpleAir sensor-network API with an OpenWeatherMap fallback, called from secure server-side Next.js API routes so keys stay hidden.
 - Estimated air quality at a chosen location using the Haversine formula and Inverse Distance Weighting, converting raw PM2.5 readings to a US EPA Air Quality Index.
- DarkOrchid — Team Software Application** Java, Android Studio, Firebase, JUnit
 - Built a full application in a team-based Agile project using object-oriented design in Java, with progress reviewed through GitHub.
 - Designed Firestore data models and wrote JUnit unit tests and UI tests to validate features.
- AI Search and Constraint Satisfaction Solvers** Python
 - Implemented AI search and constraint-satisfaction algorithms (Levin Tree Search, backtracking, CSP) with custom heuristics, evaluating decision-making and optimization strategies across problem instances.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, SQL, HTML, CSS

Front-End: React.js, Next.js, Leaflet, responsive web development, JSON

Back-End & APIs: REST APIs, Express, object-oriented programming, Node.js

Databases: SQL, SQLite, MongoDB, Firebase (Firestore), Supabase, database design, entity-relationship modeling

Tools & Cloud: Git, GitHub (branches, pull requests, code review), Azure (academic), Postman, Microsoft Office

Testing & Practices: JUnit, unit testing, UI testing, Agile/Scrum, debugging

Spoken Languages: English (fluent), French (fluent)